

Pierpaolo Vendittelli

PhD in Artificial Intelligence · Applied AI Researcher · Machine Learning Engineer

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🇮🇹 Italian citizen based in the Netherlands | 🎓 Google Scholar | ORCID

Professional Profile

Applied AI researcher and machine learning engineer with a PhD in Artificial Intelligence. I design, implement, and validate machine-learning systems for computational pathology and medical imaging, with particular depth in whole-slide imaging, segmentation, weakly supervised learning, interpretable biomarkers, and survival modelling. My work combines patient-level evaluation, external validation, scientific writing, teaching, and open research software. Recent work extends this foundation to pathology foundation models and multiple-instance learning for paediatric oncology.

Professional Experience

Postdoctoral Researcher – Medical AI and Computational Pathology

Dec. 2024 – Nov. 2026 (fixed-term)

Prinses Maxima Centre for Pediatric Oncology

Utrecht, The Netherlands

- Designing and evaluating computational-pathology pipelines for Wilms tumour whole-slide images, in collaboration with pathologists and paediatric-oncology researchers.
- Built the complete patient-level multiple-instance-learning pipeline for anaplasia classification: data splitting, model training, internal cross-validation, external validation, attention visualisation, figures, tables, and manuscript methods.
- On an external French cohort, achieved slide-level AUC 0.923 (95% CI 0.883–0.962) for anaplasia classification; all 23 anaplastic patients were identified in the evaluated cohort.
- Designed and implemented a two-stage lymph-node detection and metastasis-classification pipeline; achieved 100% node-detection recall with zero merged nodes and fully automated AUROC 0.904 in patient-grouped cross-validation.
- Retrained a 16-class Wilms tumour segmentation model and built the associated annotation-rasterisation, patient-splitting, training, and evaluation workflow.
- Provides day-to-day technical mentoring to a PhD candidate on computational-pathology experiments, model evaluation, and scientific writing; not a formally registered supervisor.

Doctoral Researcher – AI for Digital Pathology

Mar. 2021 – Feb. 2025

Radboud University Medical Center – Diagnostic Image Analysis and Computational Pathology Groups

Nijmegen, The Netherlands

- Salaried doctoral researcher in PANCAIM, an EU Horizon 2020 consortium for pancreatic-cancer AI and personalised medicine (grant agreement 101016851).
- Authored the PANCAIM Task 3.3 final deliverable and developed its underlying research projects: tumour localisation, multi-tissue segmentation, tumour-epithelium segmentation, biomarker quantification, and survival modelling from H&E whole-slide images.
- Developed an interpretable tumour-stroma-ratio workflow and led the PLOS ONE study from methodology and software through writing, submission, revisions, and reviewer response.
- Designed U-Net-family segmentation models, attention-based survival workflows, and comparative experiments between specialist multi-tissue features and the UNI pathology foundation model.
- Managed GPU/SLURM experiments, Dockerised research workflows, experiment tracking, versioned analyses, and internal/external cohort evaluation.
- Built a TeamPlay-compatible Docker container for a segmentation workflow; it was installed on the platform, with no subsequent clinical-production deployment claim.
- Taught medical-imaging and image-analysis courses and supervised two Master's research projects.

AI Engineer – Extracurricular Internship

Nov. 2020 – Feb. 2021

everis Italia S.p.A. (now NTT DATA)

Remote, Italy

- Defined intents, questions, responses, and conversational branches for a development-stage COVID-flight-refund chatbot for a major Italian airline; tested the flow in the development environment.
- Debugged and updated an automatic-subtitling workflow for English-language video on an internal social platform for a global fashion company.
- Manually annotated and quality-checked several thousand retail images with bounding boxes and labels for an object-recognition training dataset.

Research Intern

Feb. 2020 – Aug. 2020

ViCOROB – NeuroImage Computing Group, University of Girona

Girona, Spain / remote from Cassino

- Benchmarked ResU-Net, conditional GAN, and CycleGAN architectures for bidirectional T1/FLAIR MRI synthesis on WMH and BraTS 2018 datasets.
- Implemented medical-image preprocessing, model training, and quantitative evaluation using SSIM and MSE.
- Showed that ResU-Net achieved stronger SSIM and lower MSE than the GAN variants under data-scarce conditions.
- Delivered the work as the MAIA Master's thesis, graded 8.6/10 in Spain.

Mortgage Data Entry Assistant – Occasional Voucher-Based Work

2013 – 2016

Co.Da.P.S. S.r.l.

Cassino, Italy

- Read mortgage contracts and entered structured values in Excel for analyst review, including principal, rates, duration, instalments, dates, and contractual clauses.
- Completed several hundred contract records while studying; the role did not include independent calculations or financial analysis.

Research Programs and Technical Contributions

Automated Anaplasia Classification in Wilms Tumour

2025 – Present

Prinses Maxima Centre for Pediatric Oncology

Pediatric Oncology

- Designed a weakly supervised WSI pipeline combining Virchow2 foundation-model features, multiple-instance learning, patient-level cross-validation, and slide-level attention maps.
- Best internal configuration achieved mean AUC 0.871 ± 0.055 ; external French-cohort AUC was 0.923 (95% CI 0.883–0.962).
- Wrote the manuscript methods and generated all figures, tables, and attention maps; manuscript is in preparation and authorship/journal selection remain unresolved.

Automated Lymph-Node Detection and Metastasis Classification in Wilms Tumour

2025 – Present

Prinses Maxima Centre for Pediatric Oncology

Pediatric Oncology

- Conceived and implemented a two-stage detection-then-classification pipeline for post-chemotherapy Wilms tumour specimens.
- The Virchow2-feature detector achieved 122/122 node recall, zero merged nodes, and micro Dice 0.890; the fully automatic positive-versus-negative classifier reached AUROC 0.904.
- External French-cohort labels are pending; this is ongoing work and the manuscript has not yet been drafted.

Automatic Tumor Segmentation in Pancreatic Whole-Slide Images

2021 – 2022

Radboud University Medical Center

PhD Research

- Built patch-based single-task and multi-task CNN pipelines for coarse PDAC tumor localization from H&E slides.
- The auxiliary classification branch reduced false positives; the best multi-task configuration achieved median Dice 0.934 on the study cohort.
- Established the tumor-localization stage used by downstream tissue and biomarker analyses.

Automatic Tumor-Stroma-Ratio Quantification

2021 – 2024

Radboud University Medical Center

PhD Research

- Developed a multi-stage WSI system combining tumor-bulk and epithelium segmentation with IHC-guided supervision to quantify interpretable tissue biomarkers.
- Evaluated 368 patients across four datasets involving material from 25 Dutch laboratories; obtained median external Dice of 0.75 for tumor epithelium and 0.86 for tumor bulk.
- Demonstrated reproducible prognostic value for automatically computed TSR, including 6-month survival discrimination on an independent cohort.
- Led conceptualisation, methodology, software, validation, visualisation, and original manuscript drafting for the PLOS ONE study.

Multi-Biomarker and Attention-Based Survival Modeling

2024 – 2026

Radboud University Medical Center and PanGenEU collaborators

PhD Research

- Integrated tumor-stroma ratio, mitosis density, stromal cell density, and tumor-to-tissue ratio in multivariate PDAC survival analyses.
- Designed an attention-based survival pipeline agnostic to the feature extractor and compared UNI foundation-model representations with specialist multi-tissue features.
- Evaluated regression and Cox proportional-hazards objectives across RUMC, TCGA-PAAD, CPTAC-PDA, and the external PanGenEU cohort.
- Observed stronger UNI performance internally and better specialist-feature generalisation externally, highlighting domain shift, cohort size, and pretraining-overlap risks.

Multimodal Clinical AI for Pancreatic Cancer

2021 – 2025

PANCAIM consortium collaborations

International Multi-Center Research

- Contributed to multimodal prognostication combining CT imaging and clinical data and to an international paired non-inferiority study of AI and radiologists for pancreatic-cancer detection.
- Co-authored reviews and perspective work on clinical AI research priorities, multimodal fusion, implementation challenges, and translational evaluation.

Education

PhD in Artificial Intelligence

Mar. 2021 – Jun. 2026

Radboud University

Nijmegen, The Netherlands

- Dissertation: “Learning from histopathology images: AI-driven biomarkers for pancreatic ductal adenocarcinoma”.
- Public defense completed on 1 June 2026.
- Supervisors: Prof. Geert Litjens and Prof. Jeroen van der Laak; co-supervisor: Dr. John Hermans.
- Research home: Department of Radiology and Nuclear Medicine; Diagnostic Image Analysis and Computational Pathology Groups.
- Completed the Radboudumc Advanced Graduate School training program.

Erasmus Mundus Joint M.Sc. in Medical Imaging and Applications (MAIA)

University of Burgundy, University of Cassino and Southern Lazio, University of Girona

Sep. 2018 – Sep. 2020

France, Italy, Spain

- French degree: Master Sciences, Technologies, Sante – Signal and Image Processing, Computer Vision / Erasmus Mundus track; 120 ECTS; Mention Bien.
- Overall results documented as Bien in France, 102/110 in Italy, and 8.6/10 in Spain.
- Master's thesis: comparative T1/FLAIR MRI synthesis using ResU-Net, conditional GAN, and CycleGAN.
- Received a full two-year tuition-fee scholarship; living costs were self-funded.

B.Sc. in Computer and Telecommunications Engineering (L-8)

University of Cassino and Southern Lazio

Oct. 2012 – Dec. 2017

Cassino, Italy

- Final grade: 94/110.
- Thesis: "A sensor unit for in-shoe measurement"; team project on a 3D-printed sensor for diabetic-patient shear-force measurement, with emphasis on data acquisition and calibration.
- Relevant coursework included programming, object-oriented programming, databases, signal theory, telecommunications networks, electronics, control systems, and mathematical analysis.
- Erasmus+ exchange at Fontys University of Applied Sciences, the Netherlands (2016–2017), supported by an approximately EUR 200/month grant.

Italian Classical High-School Diploma (Maturita Classica)

Liceo Classico Giosue Carducci

Sep. 2007 – Jul. 2012

Cassino, Italy

Final grade: 77/100.

Detailed Coursework and Doctoral Training

Computer Vision and Image Processing Foundations

University of Burgundy – MAIA Semester 1

2018 – 2019

Dijon / Le Creusot, France

- Software Engineering – 17.68/20, 5 ECTS; Introduction to Image Processing – 16.7/20, 6 ECTS.
- Applied Mathematics – 14.7/20, 6 ECTS; Digital Signal Processing – 15/20, 6 ECTS.
- Sensors and Digitization – 14/20, 5 ECTS; Local Culture – 17.5/20, 2 ECTS.
- Semester result: 15.727/20, 30 ECTS.

Robotics, Distributed Systems, Pattern Recognition, and Advanced Imaging

University of Cassino and Southern Lazio – MAIA Semester 2

2019

Cassino, Italy

- Introduction to Robotics – 13.429/20, 5 ECTS; Parallel Processing Systems – 15/20, 6 ECTS.
- Distributed Programming and Networking – 20/20, 6 ECTS; Pattern Recognition – 16/20, 6 ECTS.
- Advanced Image Analysis – 15/20, 5 ECTS; Local Culture – 20/20, 2 ECTS.
- Semester result: 16.272/20, 30 ECTS.

Medical Image Computing and Digital Health

University of Girona – MAIA Semester 3

2019 – 2020

Girona, Spain

- Medical Image Registration and Applications – 15.6/20, 6 ECTS; Medical Image Segmentation and Applications – 15.4/20, 6 ECTS.
- Computer-Aided Surgery and Medical Robotics – 15/20, 6 ECTS; Computer-Aided Diagnosis – 16/20, 5 ECTS.
- eHealth – 16.4/20, 5 ECTS; Local Culture – 15/20, 2 ECTS.
- Semester result: 15.6/20, 30 ECTS.

Research Internship and Master's Thesis

University of Girona – MAIA Semester 4

2020

Girona, Spain / Cassino, Italy

- Research stage and thesis – 13.987/20, 30 ECTS; Spanish assessment 8.6/10.
- Final M2 admission result: 14.794/20, Mention Bien; rank documented as 15/21.
- Internship ran Feb.–Aug. 2020: in person in Girona until Apr. 2020, then remotely from Cassino because of COVID-19.

Erasmus+ Exchange Coursework – Electrical and Electronic Engineering

Fontys University of Applied Sciences

2016 – 2017

The Netherlands

- Completed 50 documented ECTS across Advanced Analog Design (8.4/10), Advanced Control Systems (8.2/10), Telecom/Wireless (6.5/10), Research Methods and Trends (7.0/10), Sensor Technology for Mechatronic Systems (6.8/10), and Project Semester 7 (6.9/10).
- Also completed Minor Electrical Magnetic Compatibility (6.8/10), Minor Integrated Product Development, Analogue Electronics 4 (6.6/10), and Telecommunication 1.

BSc Coursework – Computer and Telecommunications Engineering

2012 – 2017

University of Cassino and Southern Lazio

Cassino, Italy

- Programming and computing: Foundations of Computer Science (24/30, 9 ECTS), Object-Oriented Programming (24/30, 6 ECTS), Programming Techniques (pass, 6 ECTS), Computer Architecture (25/30, 9 ECTS), and Databases (19/30, 6 ECTS).
- Mathematics and signals: Mathematical Analysis I (20/30, 12 ECTS), II (21/30, 9 ECTS), and III (19/30, 9 ECTS); Probability and Information (19/30, 6 ECTS); Signal Theory (20/30, 9 ECTS).
- Telecommunications: Foundations of Telecommunications (25/30, 9 ECTS), Telecommunications Networks (21/30, 9 ECTS), Guided Propagation (25/30, 6 ECTS), Foundations of Electromagnetic Waves (21/30, 6 ECTS), and Microwave Apparatus Design (28/30, 6 ECTS).
- Electronics and control: Electronics (26/30, 9 ECTS), Electronic Measurements (26/30, 9 ECTS), Electrical Engineering (25/30, 12 ECTS), Foundations of Dynamic Systems (25/30, 6 ECTS), and Foundations of Automatic Control (25/30, 6 ECTS).
- Other completed coursework: General Physics (28/30, 12 ECTS), Business Economics and Organisation (26/30, 6 ECTS), and Thesis Preparation (pass, 3 ECTS).

Formal PhD Training

2021 – 2025

Radboud Institute for Health Sciences / Radboudumc

Nijmegen, The Netherlands

- RIHS Introduction Course for PhD Candidates – 15 hours (2021), including project planning, supervision, time management, scientific communication, and career development.
- Radboudumc Scientific Integrity – 20 hours (2025).
- RIHS PhD Retreat – 16 hours (2021).

Pathology of the Exocrine Pancreas – Of Men and Mice

1–3 Dec. 2022

Spanish National Cancer Research Centre (CNIO)

Madrid, Spain

- Three-day meet-the-expert international course on exocrine-pancreas pathology; attendance certificate received.

Teaching, Supervision, and Mentoring

Technical Mentoring of a PhD Candidate

2025 – Present

Prinses Maxima Centre for Pediatric Oncology

Utrecht, The Netherlands

- Provides day-to-day technical guidance on computational-pathology experiments, model evaluation, and scientific writing; not a formally registered PhD supervisor.

Teaching Assistant – Intelligent Systems for Medical Imaging

2022 and 2023

Radboud University Medical Center

Nijmegen, The Netherlands

- Two editions, 60 hours each; prepared practical exercises, ran front-facing exercise sessions, corrected assignments, and provided individual feedback by email.
- Contributed approximately 12% of lectures in the first edition and 25% in the second.

Teaching Assistant – Image Analysis

2023, 2024, and 2025

Radboud University Medical Center

Nijmegen, The Netherlands

- Three editions, 50 hours each; developed and updated notebooks and slides, led practical sessions, assessed assignments, and provided feedback.
- Topics included segmentation, filtering, convolution, morphological operations, and Hough transforms, using Python, scikit-image, and OpenCV.

Master's Thesis Supervision

2023 and 2024

Radboud University Medical Center

Nijmegen, The Netherlands

- Two documented supervision periods, 185 hours each; 370 supervision hours in total.
- Supervised Tessa de Wijs on a computational-pathology thesis and review work, and Lisa Hensens on a thesis using and extending Pierpaolo's code for experimental work.

Documented Doctoral Development Activities

2021 – 2025

Radboudumc Advanced Graduate School Portfolio

Nijmegen, The Netherlands

- Completed 803 documented hours across formal courses, conferences, teaching assistance, and Master's thesis supervision.

Peer-Reviewed Publications and Proceedings

Artificial intelligence and radiologists in pancreatic cancer detection using standard-of-care CT scans (PANORAMA): an international, paired, non-inferiority, confirmatory, observational study

Alves N, Schuurmans M, Rutkowski D, Saha A, Vendittelli P, et al. *The Lancet Oncology*, Nov. 2025.

doi:10.1016/S1470-2045(25)00567-4.

End-to-end prognostication in pancreatic cancer by multimodal deep learning: a retrospective, multicenter study

Schuurmans M, Saha A, Alves N, **Vendittelli P**, Yakar D, Sabroso-Lasa S, Xue N, Malats N, Huisman H, Hermans J, Litjens G. *European Radiology*, 2025.

PubMed 40410330.

Navigating the landscape of multimodal AI in medicine: a scoping review on technical challenges and clinical applications

Schouten D, Nicoletti G, Dille B, Chia C, **Vendittelli P**, Schuurmans M, Litjens G, Khalili N. *Medical Image Analysis*, 105:103621, Oct. 2025.

doi:10.1016/j.media.2025.103621; earlier preprint arXiv:2411.03782.

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer

Vendittelli P, Bokhorst J-M, Smeets EMM, Kryklyva V, Brosens LAA, Verbeke C, Litjens G. *PLOS ONE*, 19(5):e0301969, May 2024.

doi:10.1371/journal.pone.0301969. First and corresponding author; conceptualisation, methodology, software, validation, visualisation, original draft, submission, revisions, and reviewer response.

Artificial Intelligence in Pancreatic Ductal Adenocarcinoma Imaging: A Commentary on Potential Future Applications

Schuurmans M, Alves N, **Vendittelli P**, Huisman H, Hermans J. *Gastroenterology*, 165(2):309–316, Aug. 2023.

doi:10.1053/j.gastro.2023.04.003.

Setting the Research Agenda for Clinical Artificial Intelligence in Pancreatic Adenocarcinoma Imaging

Schuurmans M, Alves N, **Vendittelli P**, Huisman H, Hermans J. *Cancers*, 14(14):3498, Jul. 2022.

doi:10.3390/cancers14143498.

Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas

Vendittelli P, Smeets EMM, Litjens G. *SPIE Medical Imaging Proceedings*, 12039:120391F, Apr. 2022.

doi:10.1117/12.2611542; arXiv:2112.01533. First author.

Automatic quantification of TSR as a prognostic marker for pancreatic cancer

Vendittelli P, Bokhorst J-M, Smeets EMM, Kryklyva V, Brosens LAA, Verbeke C, Litjens G. *Medical Imaging with Deep Learning (MIDL)*, Sep. 2023.

Conference extended abstract and poster. First author.

Additional Research Outputs

Combining AI-based Biomarkers for PDAC Survival

Hensens L, Sabroso-Lasa S, Verbeke C, Malats N, Litjens G, **Vendittelli P**. *COMPAYL Workshop, MICCAI*, Sep. 2025.

Workshop contribution and poster on multi-biomarker survival modelling. Pierpaolo wrote the paper from the associated Master's thesis, handled submission/revisions, and presented the poster.

Comparing feature extractors for pancreatic-cancer survival from digitized histopathology

Vendittelli P and collaborators. *PhD dissertation chapter / manuscript-stage research*, 2025–2026.

Submission-ready but not yet submitted. Attention-based survival modelling comparing UNI foundation-model features with specialist multi-tissue representations across internal and external cohorts; Pierpaolo is planned first and corresponding author.

Conference Presentations and Participation

European Congress of Digital Pathology (ECDP 2026)

Jun. 2026

Poster Presenter

Graz, Austria

“Weakly Supervised Deep Learning for Anaplasia Classification in Wilms Tumor Whole-Slide Images” – Pierpaolo authored the abstract, created and presented the poster, and was the only attending author. Selected among the Top 10 posters.

COMPAYL Workshop, MICCAI 2025

Sep. 2025

Poster Presenter

Daejeon, South Korea

“Combining AI-based Biomarkers for PDAC Survival” – multi-tissue segmentation and multivariate survival prediction. Pierpaolo was the only attending author and presented the poster; conference participation documented as 40 hours.

Medical Imaging with Deep Learning (MIDL 2023)

Sep. 2023

Poster Presenter

Nashville, TN, USA

“Automatic quantification of TSR as a prognostic marker for pancreatic cancer”. First author and sole presenter; conference participation documented as 32 hours.

SPIE Medical Imaging 2022

Apr. 2022

Poster Presenter

San Diego, CA, USA

“Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas”. First author and sole attending/presenting author; conference participation documented as 40 hours.

Open-Source and Personal Engineering Projects

Pathology Segmentation Pipeline

2026

github.com/PierpaoloV/pathology-segmentation-pipeline

Open-Source Framework

- Dockerized WSI pipeline for asynchronous tile inference, tissue/background segmentation, multi-tissue classification, epithelium segmentation, and tumor-stroma-ratio computation.
- Designed for reproducible preprocessing, maintainability, and reuse; published through GitHub, Docker Hub, and a separate TeamPlay-compatible container workflow.

AI Job-Search Agent

2026

github.com/PierpaoloV/job-agent

Personal Project

- System for Gmail job-alert ingestion, provider-aware parsing, SQLite deduplication, LLM ranking against a private profile, and Telegram digest delivery.
- Includes deterministic parser/context tests and GitHub Actions support. Implemented but not yet operational on a real job-alert account.

AI Social Agent

2026

github.com/PierpaoloV/social-agent

Personal Project

- Human-in-the-loop workflow designed to monitor GitHub milestones and Telegram notes, draft professional posts, route them through approval, and publish through X.
- Implemented with a separate private state store and explicit privacy boundaries; currently paused and not operational.

Weekly AI Paper-Triage Agent

2026 – Present

github.com/PierpaoloV/weekly_ai_paper

Personal Project

- Fetches PubMed and arXiv literature, filters and deduplicates candidates, runs two LLM reviewer passes plus editorial arbitration, and emails a personalised weekly digest.
- Runs through GitHub Actions and maintains a candidate pool and sent-paper log; operational and sending real digests.

MRI Synthesis Benchmarking

2020

github.com/PierpaoloV/mri-brain-synthesis-from-scratch

Master's Research Project

ResU-Net, conditional-GAN, and CycleGAN comparison for T1/FLAIR synthesis using TensorFlow/Keras and quantitative medical-image evaluation.

Left-Ventricular Segmentation

2018 – 2019

github.com/PierpaoloV/Left-Ventricular-Segmentation

MAIA Group Project

Medical-image segmentation group project completed during the first MAIA semester. Work was performed jointly; no individual contribution is separately attributed.

Il Lungo Viaggio

2018 – Present

github.com/PierpaoloV/il-lungo-viaggio

Collaborative Creative Project

Collaboratively created narrative world and story material; Pierpaolo implements the current web prototype and Unity 2D narrative-adventure vertical slice.

Honors, Fellowships, and Grants

2026	Top 10 Poster , European Congress of Digital Pathology – “Weakly Supervised Deep Learning for Anaplasia Classification in Wilms Tumor Whole-Slide Images”	Graz, AT
2018–2020	MAIA Tuition-Fee Scholarship , Full two-year university tuition for the Erasmus Mundus Joint Master Degree in Medical Imaging and Applications (MAIA)	FR / IT / ES
2016–2017	Erasmus+ Exchange Grant , Approximately EUR 200/month for the full-year exchange at Fontys University of Applied Sciences	NL

Complete Skills Inventory

Programming Python (primary language); Bash for jobs and scripts; historical academic use of MATLAB and C++. SQL knowledge is historical and not current operational expertise.

Deep Learning PyTorch; CNNs, U-Net / U-Net++, multiple-instance learning, pathology foundation-model features, representation learning, training, inference, and evaluation. TensorFlow/Keras and MONAI: prior project experience.

Machine Learning scikit-learn, supervised and weakly supervised learning, model selection, patient-level cross-validation, leakage prevention, calibration-aware evaluation, and experimentation

Computational Pathology Whole-slide images, OpenSlide, patch extraction, tissue segmentation, MIL, foundation-model features, and tumor-microenvironment analysis

Medical Imaging MRI synthesis, image registration, segmentation, computer-aided diagnosis, quantitative image evaluation, and multimodal clinical AI

Clinical Modeling Interpretable biomarkers, risk stratification, survival regression, Cox proportional hazards, c-index, time-dependent AUC, and external validation

LLM Workflows Prompt design, LLM-based review and ranking workflows, OpenAI and Anthropic integrations, and human-approval gates for personal research/productivity agents

Engineering Docker, Git, GitHub Actions, SLURM job configuration, experiment tracking, assertions and sanity checks, reproducible pipelines, and automation

Research Infrastructure HPC, SLURM, GPU clusters, experiment tracking, Weights & Biases, multi-center workflows, and TeamPlay

Research Practice Study design, methodology, scientific writing, peer-reviewed publication, literature review, visualization, reproducibility, and supervision

Certifications and Additional Learning

Advanced Graduate School Training Program

Radboudumc Advanced Graduate School

Formal certificate awarded following completion of the doctoral training program.

Completed Jun. 2026

Nijmegen, The Netherlands

Machine Learning

Coursera – Stanford University

Individual course certificate.

Jul. 2019

Online

Deep Learning Coursework

Coursera – DeepLearning.AI

Neural Networks and Deep Learning; Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization; Structuring Machine Learning Projects.

Jul.–Sep. 2019

Online

Data Science in Python Coursework

Coursera – University of Michigan

- Introduction to Data Science in Python; Applied Plotting, Charting & Data Representation in Python.

Apr.–May 2020

Online

Medical AI and Computer-Vision Coursework

Coursera – DeepLearning.AI

AI for Medical Diagnosis; AI for Medical Prognosis; Convolutional Neural Networks; Build Basic Generative Adversarial Networks (GANs).

Apr.–Oct. 2020

Online

PyTorch Guided Project

Coursera

The PyTorch Basics You Need to Start Your ML Projects.

Mar. 2021

Online

Generative AI Coursework

Coursera – DeepLearning.AI / AWS and Google Cloud

Generative AI with Large Language Models; Gen AI: Beyond the Chatbot.

Apr. 2026

Online

Scientific Service and University Governance

Peer Reviewer

2026

COMPAYL Workshop

MICCAI workshop

Reviewed three full-paper submissions; first formal peer-reviewing experience.

Elected General Councillor, Student Council

2016 – 2017

University of Cassino and Southern Lazio

Cassino, Italy

University-wide elected representative in the university's third central governance body; participated in meetings, deliberations, votes, proposals, advisory opinions, commissions, and student representation.

Languages and Interests

Italian, English, Spanish, Dutch

Languages

Italian: native. English: professional and academic working language; IELTS Academic overall 7.0 (Feb. 2018), no component below 6.5. Spanish: basic conversational ability, acquired while living in Spain. Dutch: A1 certificate, Radboud in'to Languages (2022).

Sport, music, games, and narrative design

Interests

Former youth football left back for A.S.D. Nuova Cassino; recreational CrossFit (2018–2024); guitar and basic piano; LARP and Dungeons & Dragons player; board games, anime, manga, and collaborative narrative-game development.